

Project Initialization and Planning Phase

Date	June 28, 2026
Team ID	
Project Title	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Hydra Shield: An Intelligent Flood Prediction and Early Warning System is a Machine Learning-based application developed to predict flood risk using historical weather and environmental data. It analyzes parameters such as rainfall, water level, temperature, and humidity to generate accurate flood predictions. The system uses Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost algorithms to improve prediction accuracy. Integrated with a Flask web application, it provides timely flood warnings to support disaster preparedness and decision-making.

Project Overview	
Objective	Develop an intelligent Machine Learning system that accurately predicts flood risk using historical weather and environmental data and provides timely early warnings.
Scope	The project includes data collection, preprocessing, feature engineering, model training, model evaluation, and deployment using Flask. Multiple Machine Learning algorithms are compared to identify the best-performing model for real-time flood prediction.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing significant loss of life, property, and infrastructure. Traditional forecasting methods are often slow and less accurate, making timely evacuation and disaster preparedness difficult.
Impact	An intelligent flood prediction system enables early warning, improves disaster preparedness, supports timely decision-making, reduces economic losses, and helps protect human lives.
Proposed Solution	
Approach	Historical weather and flood datasets are preprocessed and used to train Machine Learning models including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. The best-performing model is integrated into a Flask web application using Pickle to provide real-time flood risk prediction.

Key Features	• Automated Flood Risk Prediction • High Accuracy using Machine Learning • Real-Time Prediction through Flask Web Application • User-Friendly Interface • Early Warning Support for Disaster Management • Multiple Model Comparison and Evaluation • Secure and Scalable Architecture
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Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, XGBoost, Joblib
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, Size, Format	Historical Flood Prediction Dataset (Kaggle), CSV Format, 20,000+ Weather Records